

CALIBRATION OF A PRESSURE GAGE USING DEAD-WEIGHT TESTER

AIM

To study the pressure measurement using Deadweight Tester for Bourdon–tube pressure gage.

Theory

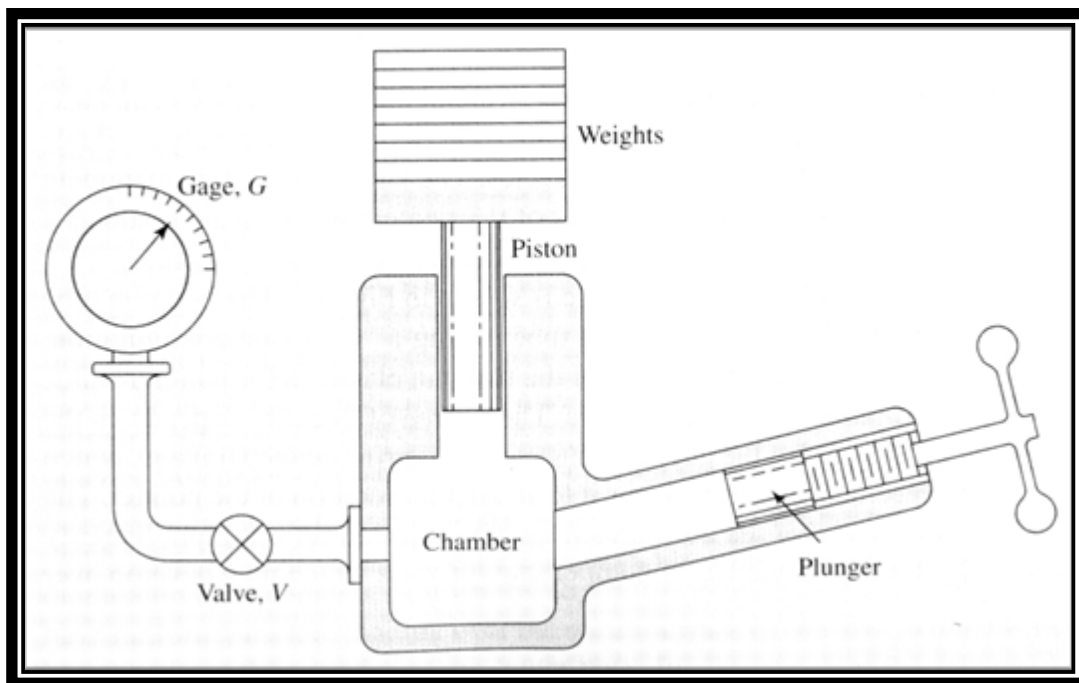


Figure 1. Deadweight Tester for Bourdon–tube pressure gage

A dead-weight tester is a device used for balancing a fluid pressure with a known weight.

Typically, it is used for static calibration of pressure gages. As seen in Figure 1, the apparatus is set up for calibration of the pressure gage G. The chamber and cylinder of the tester are filled with clean oil by first moving the plunger to its most forward position and then slowly withdrawing it while the oil is poured in through the opening for the piston. The gage to be tested is installed and piston pressure is now transmitted to the gage when the valve is opened. This pressure may be

varied by adding weights to the piston or by using different piston-cylinder combinations of varying areas. The viscous friction between the piston and the cylinder in the axial direction may be substantially reduced by rotating the piston-weight assembly while the measurement is taken. As the pressure is increased, it may be necessary to advance the plunger to account for the compression of the oil and any entrapped gasses in the apertures.

The accuracies of dead-weight tester are limited due to:

- (1) the friction between the cylinder and the piston,
- (2) the uncertainty in the area of piston,
- (3) the buoyancy error and
- (4) gravity error.

The gravity error due to elevation, z [m], and latitude, ϕ [°] can be corrected by the dimensionless correction factor as:

$$e = -(2.637 \cdot 10^{-3} \cos(2\phi) + 2.9 \cdot 10^{-8} z + 5 \cdot 10^{-5}) \text{ gravity } \phi$$

and the air buoyancy error, e_{buoyancy} , by

$$e_{\text{buoyancy}} = - \rho_{\text{air}} / \rho_{\text{massess}}$$

Then, the indicated pressure of gage, p_i , can be corrected as:

$$P = p_i (1 + e_{\text{gravity}} + e_{\text{buoyanc}})$$

2. Experiment setup

Student will perform experiment using the Dead-weight tester (Figure 1). Students are allowed to use references, handbooks, standards and manufactures catalogs. Students should also demonstrate skills in using available software.

3. Step by step procedure

- 1) Mount the pressure gage to be calibrated in its place.
- 2) Remove the cylinder tap and read the pressure gage for zero value.
- 3) Place the piston into cylinder and read the pressure gage for the pressure which is written

on the piston.

4) Add the first mass block on the cylinder and read the pressure gage. You need to advance the plunger as the weights are increased in order to insure free movement and rotation of the piston inside the cylinder

5) Repeat step 4 until all mass blocks are added (loading)

6) Tabulate the following data:

a. Experiment number,

b. The pressure gage readout, $p_{g,i}$,

c. The total pressure of the dead-weight tester (including the piston pressure), p_c ,

7) Remove the mass block which is lastly added and read the pressure gage (unloading)

8) Repeat step 7 until all mass blocks are removed

RESULT

Thus pressure measurement using Deadweight Tester for Bourdon–tube pressure gage is studied.